Class Diagram

Identity & Access Management

User
- -userId: UUID
- -emailId: String
- -fullname: String
- -contactNo: String
- -passwordHash: String
- -role: RoleType
- + findByEmail(email: String): User
- + UpdatePasswordHash(newHash: String): Void

RoleType (Enum)
- ADMIN
- MANAGER
- VOLUNTEER
- DONOR

Administrator
- -adminLevel: String

Manager
- -department: String
- +registerManager(details: String, branchId: UUID): Manager

Volunteer
- -isAvailable: Boolean
- + registerVolunteer(details: String, branchId: UUID): Volunteer
- + checkAvailability(): Boolean
- + setAvailability(status: Boolean): Void

Donor
- -donorTier: String

Security & Governance

OTPVerification
- -otpId: UUID
- -code: String
- -generatedAt: DateTime
- -isUsed: Boolean
- + generateOTP(userId: UUID): OTPVerification
- + validateCode(inputCode: String): Boolean
- + markAsUsed(): void

AuditEvent
- -eventId: UUID
- -action: String
- -timestamp: DateTime
- -status: String

Notification
- -notificationId: UUID
- -messageText: String
- -sentAt: DateTime
- + sendOTPVerification(contact: String, code: String): void

Core NGO Operations

Donation
- -donationId: UUID
- -pickupTime: DateTime
- -pickupLocation: String
- -donationCategory: DonationCategory
- -status: DonationStatus
- + CreateDonation(details: String, donorId: UUID): Donation
- + setVolunteer(volunteerId: UUID): void
- + updateStatus(newStatus: DonationStatus): Void
- + getDonationsByBranch(branchId: UUID): List<Donations>

Region
- -regionId: UUID
- -regionName: String
- + CreateRegion(name: String): Region

Branch
- -branchId: UUID
- -branchName: String
- -address: String
- + createBranch(name: String, addr: String, regId: UUID) Branch

DonationStatus (Enum)
- PENDING
- ASSIGNED
- COMPLETED

DonationCategory (Enum)
- FUNDS
- FOOD
- CLOTHING

Financial & Records

PaymentTransaction
- -transactionId: UUID
- -amount: float
- -currency: String
- -paymentMethod
- -processedAt: DateTime
- + processPayment(amount: float, method: String): PaymentTransaction

Reciept
- -reciptId: UUID
- -pdfDocumentURL: String
- -generatedAt: DateTime
- + generateReciept(txId: UUID): Reciept
- + getPdfDocumentUrl(): String

Relationships

- User 1 — requests — 0..* OTPVerification
- User 1 — generates — 0..* AuditEvent
- User 1 — receives — 0..* Notification
- Administrator → User
- Manager → User
- Volunteer → User
- Donor → User
- Manager 1 — manages — 1 Region
- Manager 1 — assigned to — 1 Branch
- Volunteer 0..* — executes — 0..* Donation
- Volunteer 1 — assigns — 0..* Donation
- Donor 1 — initiates — 1..* Donation
- Donation 1 — involves — 0..* PaymentTransaction
- PaymentTransaction 1 — generates — 1 Reciept
- Region 1 — contains — 1..* Branch
- Donation → DonationStatus
- Donation → DonationCategory